



Generation expansion planning considering health and societal damages – A simulation-based optimization approach



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ABSTRACT

Electricity generation expansion planning models determine the optimal technology-capacity-investment strategy that minimizes market costs including investment costs, and fixed and variable operating & maintenance costs over a long-term planning horizon. From a market cost perspective, fossil fuels are among the most economical sources of electricity, and thus are the primary sources of energy for electricity. However, these energy sources create by-products that have harmful health effects upon exposure. In this paper, a simulation-based, metamodeling approach is leveraged to quantify health damages associated with power grid expansion decisions by linking the outputs of generation expansion planning simulations with a screening tool that quantifies the human health damages from the electricity sector. Using this as a surrogate function for health damages, these costs are included in the objective function of a generation expansion planning model, in addition to market costs and the social damages of carbon emissions and methane leakage to minimize societal damages. Applying an improvement algorithm, candidate data points are selected to enhance metamodel prediction capability. Finally, using an updated metamodel, a new expansion plan is found. This framework enables researchers to better understand the health implications of long-term capacity expansion decisions.

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1. Introduction

Fossil fuels, mainly coal and natural gas, are the primary sources of energy for electricity in the United States, accounting for nearly 67% of electricity generation, due to the fact that they are economical and reliable energy sources [1]; [2]; [3]. However, burning fossil fuels to generate electricity produces air emissions that contribute to smog, which can ultimately lead to heart conditions, chronic lung conditions such as asthma, bronchitis, and emphysema, and other negative health implications [4]. Furthermore, these health risks are magnified when the weather is warmer and sunnier, as electricity demand is highest during these periods [5]. As a result of these health implications, additional costs are passed onto consumers, and society at large, in the form of externalities or damages [4].

Greenhouse gases, which are also by-products of fossil fuels, lead to increased global temperatures, yielding changes in precipitation, increased frequencies and intensities of extreme weather events, as well as rising sea levels [1]; [2]; [3]. These consequences impact human health by affecting our sources of food and water, air quality, and exposure to extreme weather events [1]; [2]; [3]. The most prevalent greenhouse gases in atmosphere are carbon dioxide (CO₂), accounting for approximately 81% of the total greenhouse gas emissions in the United States as of 2014, and methane (CH₄), accounting for roughly 11% of U.S. greenhouse gas emissions in 2014 [1]; [2]; [3]. Furthermore, electricity generation accounted for nearly 30% of all U.S. greenhouse gas emissions in 2014 – approximately two-thirds of which are from burning fossil fuels [1]; [2]; [3].

The U.S. Environmental Protection Agency (EPA) conducted a comprehensive study to quantify the estimated societal damages, in terms of \$/metric ton of CO₂, associated with its air emissions [1]; [2]; [3]. CH₄ is the second most prevalent greenhouse gas emitted in the United States [1]; [2]; [3]. This is mainly due to leakage from the production, processing, storage, transmission, and distribution

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of natural gas, which occurs at a rate between 1% and 8% per MWh of electricity generated [6]. According to the Intergovernmental Panel on Climate Change (IPCC), the 20-year global warming potential (or GWP) of CH₄ is approximately 72 times that of CO₂ [7]. Aside from the global warming potential of CH₄ and its approximated leakage rate, the associated health damages have not been studied in sufficient detail.

In the context of power systems planning and optimization, the topic of modeling and mitigating health damages from electricity sector air emissions is well-studied. In the short-term planning case, the Air Pollutant Optimization Model (APOM) solves the unit commitment model, for day-ahead dispatch, in the state of Georgia with the assistance of a reduced-form air quality model [8]. The objective function in the APOM tool is to minimize electricity production costs and monetized health impact cost estimates. As aligned with our research, the creators of the APOM tool recognize that health impact costs are difficult to quantify through deterministic methods, as they have a highly non-linear relationship with electricity generation and the associated air emissions. The pollutants studied in this research are ozone, PM_{2.5}, SO₂ and NO_x. To estimate health impact costs, a baseline unit commitment model along with the inputs and outputs of two EPA models are synchronized linearly to develop regression models that predict health impact costs as a function of decision variables in the unit commitment model. The first EPA model, the Community Multi-scale Air Quality model (CMAQ), geospatially and temporally models air pollutant concentrations in the atmosphere [9]. The second EPA model, the Environmental Benefits Mapping and Analysis Program (BenMAP), determines the economic value of the subsequent air pollution-related illnesses and deaths [10]. To assess the stochasticity of health impact costs, various sensitivity analyses are executed by varying key parameters in the unit commitment model, and assessing the results post-optimally in CMAQ and BenMAP.

In the context of long-term planning and capacity expansion, the most popular approach to mitigating health and other damages from air emissions is to utilize deterministic cost penalties in the objective function. Alves and Uturbey applied a single objective generation expansion planning (GEP) model with aggregate damage estimates of environmental impact costs in the objective function along with the electricity generation costs to study expansion planning decisions and the associated health damages in Brazil [11]. Similar work was done by Nguyen in Vietnam, considering health damages as deterministic penalties in the objective function of the model [12]. Over a 20-year planning horizon, their models suggest major reductions in fossil fuel dispatch in favor of significant investments in renewable energy technologies and other low carbon emitting technologies. Related research, considering expansion of both the generation and transmission networks, was done in Saudi Arabia's electricity sector, [13].

Another common approach used to develop expansion plans that mitigate health and other damages is multiple objective optimization. Karaki et al. solve a bi-objective optimization model with the goal of minimizing the competing objective functions of electricity generation costs and deterministic environmental impact costs or emissions costs [14]. As extended by Becker et al. these researchers solve a bi-objective GEP model by weighting each of these objectives a portfolio of potential expansion plans is developed [15]. Selcuklu et al. further extend this work by solving a GEP problem with the aims to minimize market costs and air pollutant emissions (and consequently adverse health effects) simultaneously in a bi-objective model. Additionally, this research introduces a Pareto Uncertainty Index (PUI) to understand the impact of uncertainty in climate change projections, thus allowing the ranking of solutions on the Pareto front based on dominance

criterion [16]. Gitizadeh et al. solve a multi-objective generation expansion planning model that considers renewable investment options and various economic and government incentive decisions. The authors simultaneously maximize economic returns and minimize fuel costs and carbon emissions [17]. Chen et al. developed a bi-objective generation expansion planning model that simultaneously minimizes total system costs and carbon emissions in China [18].

Researchers have also used GEP models to study the impact of various strategies for integrating renewable energy sources in to the electric power grid in order to reduce emissions and mitigate health damages. To study the impact on increased wind generation on capacity expansion plans, Schenk and Chan formulate the GEP model as a least-cost, mixed integer linear program with minimum wind dispatch requirements over a 19-year planning horizon [19]. Farghal and Aziz extended this research by restricting technology investments for capacity expansion to renewable sources, which yields significant emissions reductions, but increases market costs [20]. Nualhong et al. consider CO₂ emissions by considering a penalty for excessive CO₂ emissions as well as minimum investment requirements for biomass capacity [21]. Fell and Lin solve a GEP with hourly dispatch decisions and stochastic parameters to evaluate the cost effectiveness of various renewable policies, such as renewable portfolio standards, feed-in tariffs, and production subsidies. Additionally, they evaluate the impact of increased penetration of wind and solar generation in Texas on electricity prices and market dynamics [22]. To evaluate long-term decarbonization strategies for the Portuguese market, Amorim et al. employ the Integrated MARKAL-EFOM (MARKet ALlocation-Energy Flow Optimization Model), to find least-cost investment strategies through 2050 [23]. Similar work was done by Pereira et al. in the Portuguese market using a 10-year, mixed integer model determining optimal integration of renewable energy sources [24]. Seddighi and Ahmadi-Javid solve the generation expansion planning problem with transmission expansion decisions, while incorporating uncertainties about future, as well as potential disruptions. Their work also considers various sustainability regulations and policies that are evaluated for the Iranian market [25]. Koltsaklis and Georgiadis evaluate the impact of CO₂ emissions pricing in the context of GEP by including unit commitment constraints into the model [26]. Aryanpur and Shafiei investigate the economic impact of incorporating renewable energy sources to reduce carbon emissions in the Iranian market. Using the MESSAGE (Model for Energy Supply Strategy Alternative and their General Environmental Impacts) model, they evaluate various strategies, including investment plans, carbon taxes, and government incentives [27]. The MESSAGE model was also leveraged by G. and Aliakbarisani as a decision support tool to solve the GEP problem considering the cost of sustainability in Iran [28]. Li et al. solve a least-cost GEP model with stochastic climate change scenarios, with capacity expansion, electricity dispatch, and capacity retirement decision variables [29]. Using a two-stage stochastic programming approach, Li et al. evaluate the cost-risk tradeoff associated with integrating renewable portfolio standards over a 10-year planning horizon in the Guangdong Province of China [30]. Hua et al. also solve a GEP with unit commitment constraints to evaluate operational flexibility in Texas with increased renewables penetration [31].

Based on the GEP literature, there is a vast array of material related to mitigating health damages using multi-objective models, emissions limits, deterministic penalties in the objective function for air emissions, and renewable portfolio standards. However, the literature is devoid of a method that explicitly considers human health externalities in the objective function of a GEP model. The primary motivation of this research is to establish a closed-form

relationship between power grid expansion decisions and the associated health damages, in order to determine an optimal expansion plan to mitigate health risks. In this paper, these risks are addressed by presenting a framework to solve the long-term power grid capacity expansion planning problem, known as generation expansion planning (GEP). In this model, the optimal technology capacity investment strategy that minimizes health damages from air emissions, market costs including investment costs, and fixed and variable O&M costs of generation, as well as other societal damages including the social cost of carbon emissions and methane leakage is determined.

To include health damages in our model, we leverage the framework developed by Rodgers et al., which uses a metamodel to quantify health damages from NO_x and SO_2 emissions associated with power grid capacity expansion decisions, is leveraged [32,33]. In this work, health damages from simulated GEP expansion plans are evaluated in the Co-Benefits Risk Assessment (COBRA) model. COBRA is an EPA screening model that helps state and local governments assess human health damages from air emissions at the county, state, regional, or national level [34]. From these simulations, the authors derive a closed-form relationship that approximates health damages as a function of GEP decisions. This metamodel is included in the objective function of our model as a surrogate for health damages.

Societal damages from CO_2 emissions are included in the objective function by leveraging the data from EPA studies on the social costs of CO_2 emissions [1]; [2]; [3]. Furthermore, since health damages from CH_4 leakage have not been studied in sufficient detail in the literature, a methane leakage rate is approximated deterministically, and incorporated into the GEP model as an input parameter. Finally, since a portion of the GEP objective function is a model, an iterative procedure to conduct additional computer-based GEP simulations to refit our metamodel and update our GEP model solutions is presented.

The structure of the remainder of the paper is as follows. Section 2 presents the framework for developing GEP expansion plans that minimize societal damages. Section 3 demonstrates the model with a numerical example, using the Northeastern United States power grid as a case study. Finally, the paper is concluded by summarizing the main findings, as well as presenting potential research extensions of this work in Section 4.

2. Model development

In this section, the simulation-based optimization framework

used to solve our GEP model is presented. The flow chart in Fig. 1, adapted from research of Han & Zhang, summarizes the steps to create a final expansion plan.

The first step in this process, seed generation, generates a diverse sample space of data points. To accomplish this, 100 market costs minimization GEP model simulation trials are executed. Within each trial, the following model parameters are sampled from a normal distribution a coefficient of variation following a uniform random variable on the interval [0,1] [33]:

- | | |
|--|--|
| • Demand; | • Emissions limits; |
| • Peak demand; | • Costs (Investment, Fixed, and Variable); |
| • Reserve margin; | • Steam revenue; |
| • Minimum and maximum limits; | • Transmission capacity; |
| • Minimum renewable generation levels; | • Capacity factors for non-dispatchable units; and |
| • Emissions rates | • Derating values. |

Each of these simulation trials are then compared to a baseline GEP model to obtain the percent change in NO_x and SO_2 emissions from the baseline. For each trial, these values are used as inputs to be evaluated in the COBRA model to quantify health damages. To elaborate, COBRA users generate scenarios by specifying changes in the emissions, namely NO_x and SO_2 , on an annual basis for a specified region [34]. The model then quantifies the associated changes in changes in particulate matter ($\text{PM}_{2.5}$) concentrations between a baseline scenario and an experimental (or control) scenario. A source-receptor matrix translates the air pollution emissions changes into changes in ambient $\text{PM}_{2.5}$. Thus, since COBRA approximates changes in $\text{PM}_{2.5}$ concentrations as part of the approximation of health damages, we do not explicitly consider this pollutant in our model.

Using a kriging metamodel, a closed-form function that relates changes in dispatching decisions from GEP model outputs to health damages from NO_x and SO_2 emissions is developed. Using this as a surrogate function for health damages, GEP model is solved. To ensure the prediction accuracy of the metamodel, and consequently yield better GEP model outputs, the expected improvement of the metamodel is evaluated to determine when to terminate the algorithm or to execute additional GEP simulations to enhance the metamodel.

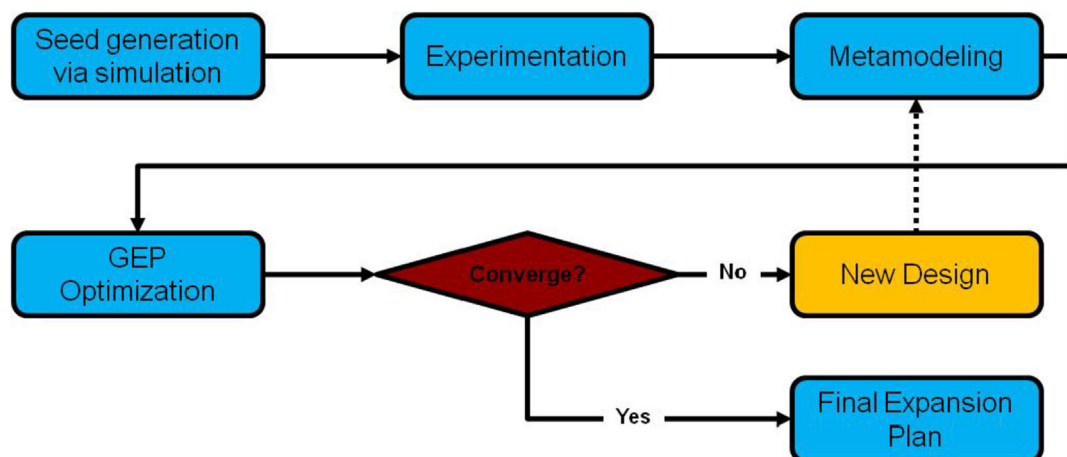


Fig. 1. Simulation-Based Optimization Flow Chart [35].

2.1. Mathematical formulation

The GEP model formulation presented in this section is the foundation of the optimization activities to be executed as part of the presented framework. This is a multi-period, multi-region GEP model, where all future costs considered in the objective function are discounted to the present time using a discount rate r to yield a net present value (NPV). Model sets, parameters, and decision variables are given in Tables 1 through 3.

Components of the objective function are given in Equations (1) through (8).

$$O_I = \sum_{y \in Y} \frac{1}{(1+r)^y} \sum_{r_1 \in R} \sum_{i \in I} \alpha_{yi} y_{yr_1i} \quad (1)$$

$$O_M = \sum_{y \in Y} \frac{1}{(1+r)^y} \left[\sum_{r_1 \in R} \sum_{i \in I} \left(\sum_{u(\forall u \leq y) \in Y} \beta_{ui} y_{ur_1i} \right) + \sum_{r_1 \in R} \sum_{i \in I^*} \beta_{yi} q_{r_1i} \right] \quad (2)$$

$$O_G = \sum_{y \in Y} \frac{1}{(1+r)^y} \sum_{t \in T} \sum_{r_1 \in R} \sum_{i \in I} v_{yti} x_{ytr_1i} \quad (3)$$

$$O_R = \sum_{y \in Y} \frac{1}{(1+r)^y} \sum_{t \in T} \sum_{r_1 \in R} \sum_{i \in I_{REV}} \mu_{yi} x_{ytr_1i} \quad (4)$$

$$O_C = \sum_{y \in Y} \frac{1}{(1+r)^y} \sum_{t \in T} \sum_{r_1 \in R} \sum_{i \in I_L} \sum_{e \in E_c} c_y x_{ytr_1i} \rho_{ie} \quad (5)$$

$$O_L = \sum_{y \in Y} \frac{1}{(1+r)^y} \sum_{t \in T} \sum_{r_1 \in R} \sum_{i \in I_L} \sum_{e \in E_c} c_y x_{ytr_1i} \hat{\lambda}_{ie} \quad (6)$$

$$\hat{O}_H = \sum_{y \in Y} \frac{1}{(1+r)^y} \sum_{t \in T} \sum_{r_1 \in R} \sum_{i \in I} \hat{\gamma}_{ytr_1i} x_{ytr_1i} \quad (7)$$

Investment costs, given in Equation (1), are the capital costs associated with capacity expansion decisions. Capacity investments of a generation unit type i in year y in region r_1 are given by the continuous variable y_{yr_1i} in units of MW. Additionally, α_{yi} is investment cost in \$/MW of unit generation unit type i in year y , r is the interest rate, Y is the set of all years in the planning horizon, and I is the set of all generating unit options. Fixed operating and maintenance costs are given in Equation (2). β_{yi} is the maintenance and operational cost (\$/MW) for unit i in year y , I^* is the set of all existing generating unit options, and q_{r_1i} is the initial capacity of unit i in region r_1 . Variable operating and maintenance costs (including fuel) are given in Equation (3), and are a function of the amount of aggregate energy generation dispatched from unit i in region r_1 period t of year y , given by x_{ytr_1i} (in MWh). The associated variable cost per MWh is given by v_{yti} . We define T to be the set of all time-periods in the model, which include: spring/fall-offpeak, spring/fall-peak, summer-offpeak, summer-peak, winter-offpeak, and winter-peak. Steam generation (or co-generation) revenue, given by Equation (4), is a function of the amount of aggregate electricity dispatched from combined heat and power units. μ_{yi} is the revenue from these units (\$/MWh). Total market costs across the network are given by the sum of all investment, fixed, and variable operating maintenance costs (Equations (1) through (3)) minus steam generation revenue (Equation (4)).

The societal damages from CO₂ emissions are approximate health and economic damages costs per ton of CO₂ emissions from electricity generation developed by the EPA [1]; [2]; [3]. These costs are included as deterministic elements in our model as given in Equation (5). These costs are given by c_y , and ρ_{ie} denotes the CO₂ emissions rate of unit i . Additionally, the societal damages of CH₄ leakage from the production, processing, storage, transmission, and distribution of natural gas are given in Equation (6) [1]; [2]; [3]. Since CH₄ leakage have not been studied in sufficient detail in the literature leakage rates are difficult to quantify, the methane leakage rate, $\hat{\lambda}_{ie}$, is calculated using the following structured

Table 1
GEP model set definitions.

Set	Definition
Y	Years in the planning horizon (2015 through 2040) – indexed by y (or u)
T	Periods in the planning horizon (summer – peak/offpeak, winter – peak/offpeak, spring/fall – peak/offpeak) – indexed by t
I	All generating units (nuclear, combined-cycle gas turbine, natural gas turbine, wind (land and offshore), biomass, coal, combined heat and power, solar, petroleum, and hydro)
I^*	All units with existing capacity in the network – indexed by i
I_R	Renewable units (biomass, wind, and solar) – indexed by i
I_L	Units with methane leak rates considered (only CCGT and natural gas turbines) – indexed by i
I_{MAX}	Units with construction limits – indexed by i
I_{REV}	Units with steam revenue (combined heat and power only) – indexed by i
I_{ND}	Non-dispatchable units (wind and solar) – indexed by i
R	Regions within the Northeastern US network (New England, NY, NYC, NJ, MD/DE, and Rest of PJM) – indexed by r_1 (or r_2)
R^*	Regional Greenhouse Gas Initiative (RGGI) subregions (New England, NY, NYC, NJ, MD/DE) within the Northeastern US network – indexed by r_1 (or r_2)
E	Emissions considered in the GEP model (NO _x , SO ₂ , CO ₂ , CH ₄) – indexed by e
E_C	Emissions considered in the GEP model to compute the social cost of carbon (CO ₂) – indexed by e
E_T	Emissions considered in the GEP model for specific time periods (SO ₂) – indexed by e
E_R	Emissions considered in the GEP model for specific regions – indexed by e
E_{MAX}	Emissions considered in the GEP model with maximum limits for all regions (NO _x only) – indexed by e
E^*	Emissions considered in the GEP model specific to RGGI regions (CO ₂) – indexed by e
$A_{(t,e)}$	Set of emissions considered by time period – indexed by (t,e)
$\Omega_{(y,r_1,r_2)}$	Set of transmission lines by year – indexed by (y,r_1,r_2)
$\Phi_{(r_1,r_2)}$	Network of regions for renewable trading – indexed by (r_1,r_2)
$\Theta_{(r_1,i)}$	Set of all renewable units within each region – indexed by (r_1,i)

Table 2
GEP model parameter definitions.

Parameter	Definition	Equations
r	Interest rate (3%)	(1)–(7)
α_{yi}	Investment costs in year y of unit i (\$/MW)	(1)
$q_{r_1,i}$	Initial capacity of unit i in region r_1 (MW)	(2), (11), (12), (15)
β_{yi}	Fixed cost (\$/MW)	(2)
v_{yti}	Variable costs in year y in region r_1 from unit i (\$/MWh)	(3)
μ_{yi}	Steam Revenue in year y from unit i (\$/MWh)	(4)
$\rho_{i,e}$	Emissions of unit i of pollutant e (lbs/MWh)	(5), (18)–(21)
c_y	Societal damages from CO ₂ emissions in year y (\$/ton)	(5), (6)
$\hat{\lambda}_{i,e}$	Approximated CH ₄ leakage rate by unit i (CO ₂ e/MWh)	(6)
$\hat{\gamma}_{yr,i}$	Approximated health damages from NO _x and SO ₂ emissions in year y in region r_1 from unit i (\$/MWh)	(7), (8)
d_{ytr_1}	Demand in period t of year y in region r_1 (MWh)	(10), (16), (17)
l_t	Transmission losses in period t (%)	(10)
h_t	Hours in period t (hours)	(11)–(14)
δ_{ti}	Derating value by season and unit (%)	(11), (12)
$\eta_{tr,i}$	Capacity factor in period t in region r_1 for non-dispatchable unit i (%)	(12)
χ_{yr,r_2}	Transmission capacity in year y from regions r_1 to r_2 (MWh)	(13), (14)
θ_i	Capacity value of unit i (%)	(15)
$d_{yr_1}^*$	Peak demand in year y in region r_1 (MW)	(15)
m_{r_1}	Reserve margin in region r_1 (%)	(15)
$\psi_{yr,i}$	Minimum renewable requirement year y in region r_1 from unit i (%)	(16)
$\psi_{yr_1}^*$	Total minimum renewable requirement in year y in region r_1 (%)	(17)
ε_{ye}^{RGGI}	RGGI emissions limits in year y of pollutant e (lbs)	(18)
$\pi_{yr_1}^*$	RGGI emissions limits in year y in region r_1 (lbs)	(19)
π_{yr_1}	Emissions limits in year y in region r_1 (lbs)	(20)
ε_{ye}	Northeastern US emissions limits in year y of pollutant e (lbs)	(21)
$k_{y,i}$	Construction limits in year y of unit i (MW)	(22)
k_i^*	Total construction limits of unit i (MW)	(23)

Table 3
GEP model decision variables.

Notation	Definition	Equations
$y_{yr,i}$	Capacity investment in year y in region r_1 of unit i (MW)	(1), (2), (11), (12), (15), (22)–(24)
$x_{ytr,i}$	Generation in period t of year y of unit i in region r_1 (MWh)	(3)–(8), (10)–(12), (16)–(21), (23), (24)
w_{ytr,r_2}	Transmission of electricity in period t of year y from region r_1 to region r_2 (MWh)	(10), (13), (14), (24)

approach [32]:

- **Step 1:** Natural gas leaks at an estimated rate of approximately 4.5% [6]. This value is denoted as a .
- **Step 2:** Approximately 3448.3 cubic feet of natural gas is used to generate 1 MWh of electricity from natural gas turbine, and approximately 2664.6 cubic feet of natural gas is used to generate 1 MWh of electricity from combined-cycle gas turbines [3]. These values are denoted using b_i , where i indexes the units with natural gas fuel sources.
- **Step 3:** Natural gas is approximately 90% methane by composition, denoted by g [3].
- **Step 4:** The value GWP is defined as the 100-year global warming potential (or GWP) of methane, which is 28 times that of CO₂ [36].
- **Step 5:** Finally, per the ideal gas law, the density of methane is a function of temperature. Based on climate change projections, temperatures are expected to increase by approximately 4°F by the year 2100 [3]. Assuming 1 atm of pressure, and prorating the temperature growth to the time-horizon considered in our GEP model, the density of methane is simulated by applying a uniform distribution to the temperature within the region of the prorated projected boundaries. The density of methane is denoted by η .
- **Step 6:** The product of a , b_i , g , GWP, and η yields the approximated CH₄ leakage rate, $\hat{\lambda}_{i,e}$, to be used in the model.

Surrogate function for total health damages from NO_x and SO₂ is given in Equation (7). Using the framework presented by Rodgers et al., we obtain the kriging metamodel in Equation (8) to compute health damages in units of \$/MWh [32,33]. This kriging interpolation model is a geospatial estimation method that predicts the values of a random field at unobserved locations based on an interpolated function of observed samples.

$$\hat{\gamma}_{yr,i}(\Delta \mathbf{x}_{yr,i}) = \mu(\Delta \mathbf{x}_{yr,i}) + \sum_{k=1}^{n(\Delta \mathbf{x}_{yr,i})} \omega_k \left[\gamma_{yi}(\Delta \mathbf{x}_{yr,ik}) - \mu(\Delta \mathbf{x}_{yr,ik}) \right] \quad (8)$$

$\hat{\gamma}_{yr,i}(\Delta \mathbf{x}_{yr,i})$ is the predicted value of human health damages (\$/MWh) in year y for unit i in region r_1 , which is a function of a vector of unobserved dispatch values, $\Delta \mathbf{x}_{yr,i}$. $\Delta \mathbf{x}_{yr,ik}$ is a vector of observed dispatch deviations obtained from the experimental procedure, indexed by k . The observed value of $\Delta \mathbf{x}_{yr,ik}$ is given by $\hat{\gamma}_{yr,i}(\Delta \mathbf{x}_{yr,i})$, and $n(\Delta \mathbf{x}_{yr,i})$ is the number of nearest neighbors to consider in the model. Kriging weight, ω_k , is derived from the residuals, and the means of the predicted response and observed response are given by $\mu(\Delta \mathbf{x}_{yr,i})$ and $\mu(\Delta \mathbf{x}_{yr,ik})$ respectively. The resulting objective function is given in Equation (9)

$$z = O_I + O_M + O_G - O_R + O_C + O_L + \hat{O}_H \quad (9)$$

The full formulation of our GEP model is given as follows:

minz

s.t.

$$\sum_{r_1 \in \Omega_{(y,r_1,r_2)}} l_t w_{ytr_1r_2} - \sum_{r_1 \in \Omega_{(y,r_1,r_2)}} w_{ytr_2r_1} + \sum_{i \in I} x_{ytr_1i} = d_{ytr_1} \quad \forall y \in Y, t \in T, r_2 \in R \quad (10)$$

$$x_{ytr_1i} \leq \left(q_{r_1i} + \sum_{u(\forall u \leq y) \in Y} y_{ur_1i} \right) \delta_{ti} h_t \quad \forall y \in Y, t \in T, r_1 \in R, i \notin I_{ND} \quad (11)$$

$$x_{ytr_1i} \leq \left(q_{r_1i} + \sum_{u(\forall u \leq y) \in Y} y_{ur_1i} \right) \delta_{ti} h_t \eta_{tr_1i} \quad \forall y \in Y, t \in T, r_1 \in R, i \in I_{ND} \quad (12)$$

$$w_{ytr_1r_2} - w_{ytr_2r_1} \leq \chi_{ytr_1r_2} h_t \quad \forall (y, r_1, r_2) \in \Omega_{(y,r_1,r_2)}, Y, t \in T, i \in I \quad (13)$$

$$w_{ytr_2r_1} - w_{ytr_1r_2} \leq \chi_{ytr_2r_1} h_t \quad \forall (y, r_1, r_2) \in \Omega_{(y,r_1,r_2)}, Y, t \in T, i \in I \quad (14)$$

$$\sum_{i \notin I_{ND}} q_{r_1i} + \sum_{u(\forall u \leq y) \in Y, i \notin I_{ND}} y_{ur_1i} + \sum_{i \in I_{ND}} q_{r_1i} \theta_i + \sum_{u(\forall u \leq y) \in Y, i \in I_{ND}} y_{ur_1i} \theta_i \geq d_{ytr_1}^* m_{r_1} \quad \forall y \in Y, r_1 \in R \quad (15)$$

$$\sum_{t \in T, r_2 \in \Phi_{(r_1,r_2)}} x_{ytr_2i} \geq \psi_{ytr_1} \sum_{t \in T} d_{ytr_1} \quad \forall y \in Y, (r_1, i) \in \Theta_{(r_1,i)} \quad (16)$$

$$\sum_{t \in T} \sum_{(r_2) \in \Theta_{(r_1,i)}, r_2 \in \Phi_{(r_1,r_2)}} x_{ytr_2i} \geq \psi_{ytr_1}^* \sum_{t \in T} d_{ytr_1} \quad \forall y \in Y, r_1 \in R \quad (17)$$

$$\sum_{t \in T} \sum_{r_1 \in R^*} \sum_{i \in I} x_{ytr_1i} \rho_{i,e} \leq \xi_{ye}^{RGGI} \quad \forall y \in Y, e \in E^* \quad (18)$$

$$\sum_{t \in T} \sum_{i \in I} x_{ytr_1i} \rho_{i,e} \leq \pi_{ytr_1} \quad \forall y \in Y, r_1 \in R, e \in E_R \quad (20)$$

$$\sum_{t \in T} \sum_{r_1 \in R} \sum_{i \in I} x_{ytr_1i} \rho_{i,e} \leq \xi_{ye} \quad \forall y \in Y, e \in E \quad (21)$$

$$\sum_{r_1 \in R} y_{ytr_1i} \leq k_{y,i} \quad \forall y \in Y, i \in I_{MAX} \quad (22)$$

$$\sum_{y \in Y} \sum_{r_1 \in R} y_{ytr_1i} \leq k_i^* \quad i \in I_{MAX} \quad (23)$$

$$x_{ytr_1i} \geq 0, w_{ytr_1r_2} \geq 0, y_{ytr_1i} \geq 0 \quad \forall (y, r_1, r_2) \in \Omega_{(y,r_1,r_2)}, Y, t \in T, i \in I \quad (24)$$

Equation (10) is the demand (or energy balance) constraint, which specifies that, for each period in a year, total generation from both existing and new generating units should be at least as much as the corresponding demand in that region. Equations (11) and (12) are related capacity constraints for dispatchable and non-dispatchable units. Equations (13) and (14) are the transmission constraints. Equation (15) is the reserve margin constraint for peak demand periods. Equations (16) and (17) are minimum renewables requirements. Equations (18) through (21) are the emissions constraints. Equations (22) and (23) are investment limit constraints on an annual basis and throughout the time horizon of the model respectively. Equation (24) specifies non-negativity for generation, transmission, and investment decision variables.

2.2. Solution procedure

Since a portion of the objective function in this GEP model is a surrogate function, it is necessary to ensure that the metamodel, and thus the resulting expansion plan, is reflective of the true relationship between inputs and outputs [35]. In our case, we consider the relationship between electricity generation and health damages. The first step in this procedure is to solve the GEP model with the surrogate function for health damages in the objective function. Secondly, using a multistart heuristic algorithm, an expected improvement (EI) problem is solved, which determines additional experimental GEP simulations to compare against the baseline GEP model for evaluation in the COBRA tool [37]. Upon experimentation in COBRA, this collection of data points is then appended to the original experimental data set, and the metamodel is refit. Equation (25) defines the expected improvement function, $E[I(\mathbf{x})]$.

$$E[I(\mathbf{x})] = \begin{cases} \left(O_{H,min} - \hat{O}_H(\mathbf{x}) \right) \Phi \left(\frac{O_{H,min} - \hat{O}_H(\mathbf{x})}{\hat{s}(\mathbf{x})} \right) + \hat{s}(\mathbf{x}) \varphi \left(\frac{O_{H,min} - \hat{O}_H(\mathbf{x})}{\hat{s}(\mathbf{x})} \right), & \text{if } \hat{s} \geq 0 \\ 0 & \text{if } \hat{s} = 0 \end{cases} \quad (25)$$

$$\sum_{(t,e) \in \Lambda_{(t,e)}} \sum_{i \in I} x_{ytr_1i} \rho_{i,e} \leq \pi_{ytr_1}^* \quad \forall y \in Y, r_1 \in R, e \in E_T \quad (19)$$

In Equation (25), $\mathbf{x}$ is a vector of dispatching decisions, each of which are given by x_{ytr_1i} . $O_{H,min}$ is the minimum value of the total COBRA health damages from all GEP model trials. $\hat{O}_H(\mathbf{x})$ is the current metamodel, and $\hat{s}$ is the model error. $\Phi(\bullet)$ and $\varphi(\bullet)$

represent the cumulative distribution function and probability density function of a standard normal distribution respectively. The formulation for the EI optimization model is given as follows:

$$\max z = E[I(\mathbf{x})]$$

s.t.

$$\sum_{r_1 \in \Omega_{(y,r_1,r_2)}} l_t w_{ytr_1r_2} - \sum_{r_1 \in \Omega_{(y,r_1,r_2)}} w_{ytr_2r_1} + \sum_{i \in I} x_{ytr_1i} = d_{ytr_1} \quad \forall y \in Y, t \in T, r_2 \in R \quad (26)$$

$$w_{ytr_1r_2} - w_{ytr_2r_1} \leq \chi_{ytr_1r_2} h_t \quad \forall (y, r_1, r_2) \in \Omega_{(y,r_1,r_2)}, Y, t \in T, i \in I \quad (27)$$

$$w_{ytr_2r_1} - w_{ytr_1r_2} \leq \chi_{ytr_2r_1} h_t \quad \forall (y, r_1, r_2) \in \Omega_{(y,r_1,r_2)}, Y, t \in T, i \in I \quad (28)$$

$$x_{ytr_1i} \geq 0, w_{ytr_1r_2} \geq 0 \quad \forall (y, r_1, r_2) \in \Omega_{(y,r_1,r_2)}, Y, t \in T, i \in I \quad (29)$$

The goal of this model is to determine the dispatch plan that maximizes the expected improvement of the metamodel. Equation (26) is the energy balance constraint, and Equations (27) and (28)

are the transmission constraints, all of which are from our baseline GEP model. The decision variables in this model are x_{ytr_1i} and $w_{ytr_1r_2}$, which are the respective dispatching and transmission decision variables. To solve this problem, we apply the previously defined multistart heuristic algorithm.

We follow the procedures outlined in this section iteratively until we reach one of the two termination criteria:

- **Case 1:** When the expected improvement model reaches zero, we terminate our model. This implies that our metamodel cannot be improved any further with more experimental data;
- **Case 2:** From iteration to iteration, if the objective function changes less than a pre-specified tolerance range, we terminate our model.

3. Numerical example

In this section, our framework is applied to a series of scenarios to evaluate the impact of constraints and objective function costs on GEP expansion plans. For these examples, we consider the PJM Interconnection LLC, which is a regional transmission organization that is part of the Eastern Interconnection grid in the Northeastern U.S., as shown in the schematic representation in Fig. 2.

The network shown in Fig. 2 has 6 regions, with the notation

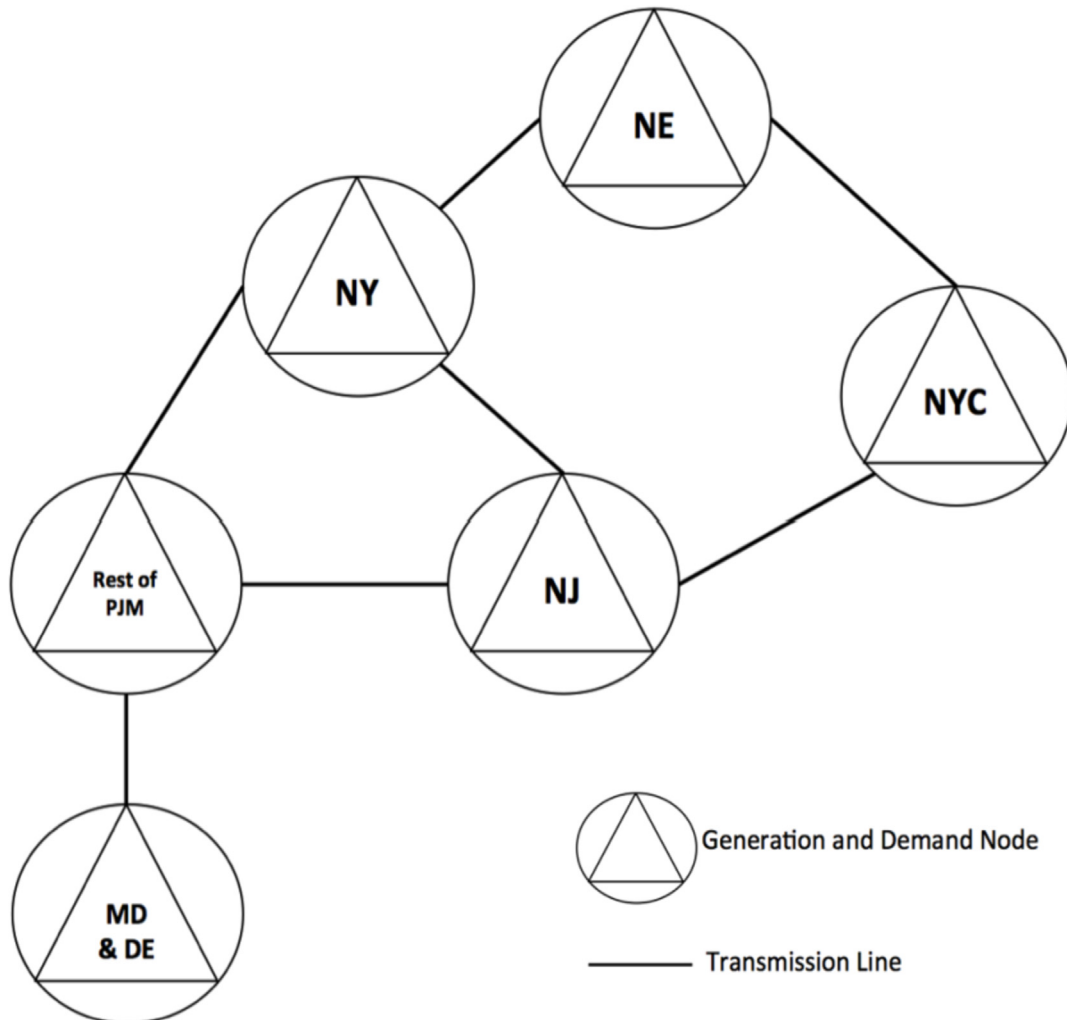


Fig. 2. Transmission network of the Northeastern United States.

Table 4

Generating units and their costs, capacities, and emissions rates.

	^a Aggregate Capacity (MW)	^b Investment Costs (\$/MW)	^b Maintenance Costs (\$/MW)	^b Variable Costs (including Fuel) (\$/MWh)	^b Fuel Costs (\$/MWh)	^b Steam Revenue (\$/MWh)	^c Human Health Externalities (\$/MWh)	^d Social Cost of CO ₂ (\$/MT of CO ₂ emitted)	^e CO ₂ (lbs/MWh)	^e NO _x (lbs/MWh)	^e SO ₂ (lbs/MWh)
Biomass	0	\$4,359,382	\$74,627	\$30.29	\$0.00	\$0.00	\$0.00	\$36	0	4	10
Combined Cycle Gas Turbine	21,828	\$1,096,358	\$13,546	\$28.02	\$15.69	\$0.00	\$4.28		1135	2	0
Combined Heat & Power	0	\$1,744,772	\$18,563	\$75.86	\$14.30	\$56.63	\$0.00		1135	2	0
Coal	59,968	\$2,713,713	\$38,327	\$25.75	\$12.91	\$0.00	\$44.58		2249	6	13
Natural Gas Turbine	16,172	\$733,926	\$12,198	\$52.30	\$15.69	\$0.00	\$4.28		1135	2	0
Hydro	24,082	\$2,909,653	\$12,198	\$0.00	\$0.00	\$0.00	\$0.00		0	0	0
Nuclear	27,799	\$4,326,537	\$104,245	\$6.35	\$5.21	\$0.00	\$2.54		0	0	0
Petroleum	4703	\$1,114,480	\$14,452	\$99.31	\$64.87	\$0.00	\$9.27		1672	4	12
Solar	446	\$6,989,283	\$13,523	\$0.00	\$0.00	\$0.00	\$1.50		0	0	0
Off-Shore Wind	0	\$4,459,051	\$98,446	\$0.00	\$0.00	\$0.00	\$0.00		0	0	0
Wind On Land	1985	\$2,226,694	\$35,088	\$0.00	\$0.00	\$0.00	\$0.00		0	0	0

^a [38].^b [39].^c [4].^d [1]; [2] [3].^e [40].

given as follows:

- NE: New England (Maine, Vermont, New Hampshire, Massachusetts, Connecticut and Rhode Island);
- NY: New York State (excluding New York City);
- NYC: New York City;
- NJ: New Jersey;
- MD & DE: Maryland, Delaware, and the District of Columbia;
- Rest of PJM: Illinois, Indiana, Kentucky, Michigan, North Carolina, Ohio, Tennessee, Virginia, and West Virginia.

Each region in the network is a source of demand, and also is capable of generating electricity. Available generation technologies to be considered in the model are combined cycle gas turbines, coal, natural gas turbines, hydro, nuclear, petroleum, solar, biomass, on-shore wind, and off-shore wind. Additionally, the configuration of the regions defines the transmission network.

The cost and emissions assumptions given in Table 4 are applied to the network presented in Fig. 2 to solve a GEP model over a 25-year time horizon (2015 through 2040). Six test cases presented in Table 5 are evaluated. Cases 1.1 through 1.3 are market cost minimization models. Cases 1.1 and 1.2, in particular, are similar to traditional, least-cost GEP models. Case 1.3 extends Case 1.2 to

include emissions limits and renewable portfolio standards as constraints. Cases 1.4 and 1.5 extend Case 1.2 to include health damages in the objective function. Case 1.6 is an extension of Case 1.5, which applies the simulation-based optimization approach discussed in the previous section. Cases 1.1 through 1.5, however, employ a CPLEX linear programming (LP) solver to obtain an optimal solution.

Fig. 3 displays annual dispatching results for each case. In the Baseline and Reserve Margin cases (Cases 1.1 and 1.2 respectively), approximately 80% of total generation is from nuclear, coal, and natural gas sources, which is typical in models with such formulations. Implementing emissions limits and renewable portfolio standards as constraints in the model reduces fossil fuel generation (Case 1.3) and increases combined heat and power as well as renewable generation from wind on land, solar, and biomass units. However, in cases where health damages are included in the objective function, coal generation is significantly displaced by other units. In the Social Cost of CO₂ and CH₄ Leakage case (Case 1.4), coal generation is displaced by natural gas and combined cycle gas turbines. Including deterministic health damages in the objective function leads to similar results (Case 1.5), and also proposes dispatch from off-shore wind units. Replacing these deterministic damages with simulated health damages (Case 1.6) yields

Table 5

Full model test cases.

	Objective Function Costs						Constraints					
	Market Costs			Health Damages								
	Investment	Fixed O&M	Variable O&M (including fuel)	Social Cost of CO ₂ and CH ₄ Leakage	Deterministic NO _x and SO ₂ Multipliers	Simulated Human Health Externalities (COBRA)	Energy Balance	Capacity	Construction Limits	Reserve Margin	Emissions Limits	Renewable Portfolio Standards (RPS)
Case 1.1 - Baseline	✓	✓	✓				✓	✓	✓			
Case 1.2 - Reserve Margin	✓	✓	✓				✓	✓	✓	✓		
Case 1.3 - Emissions & RPS Constraints	✓	✓	✓				✓	✓	✓	✓	✓	✓
Case 1.4 - Social Cost of CO ₂ and CH ₄ Leakage	✓	✓	✓	✓			✓	✓	✓	✓		
Case 1.5 - Deterministic Health Damages	✓	✓	✓	✓	✓		✓	✓	✓	✓		
Case 1.6 - Simulated Health Damages	✓	✓	✓	✓		✓	✓	✓	✓	✓		

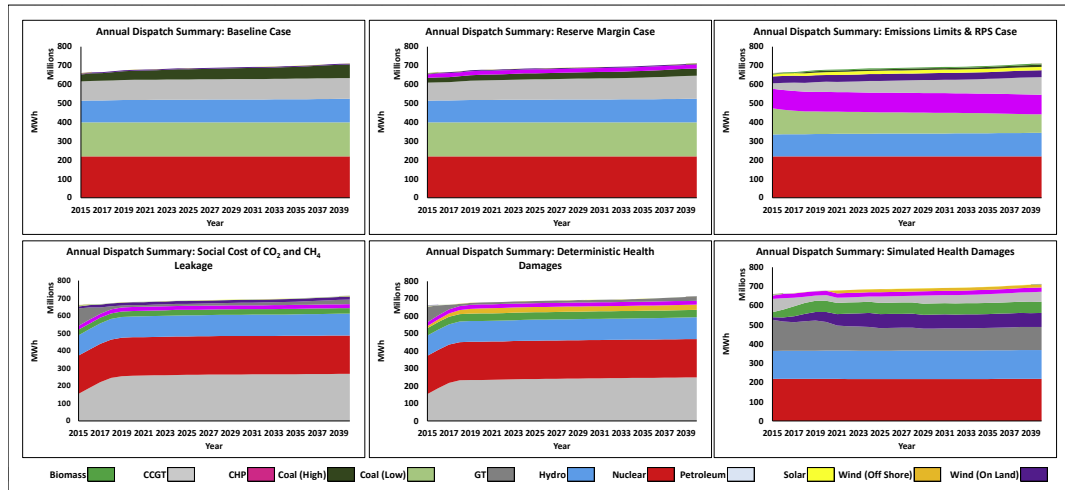


Fig. 3. Annual dispatch summary dashboard.

Table 6
Summary of new investments.

Case	Capacity Investments (MW)									
	Wind (On Land)	CCGT	CHP	Solar	Petroleum	Biomass	GT	Wind (Off Shore)	Coal	Hydro
Case 1.1 – Baseline	–	1629	2568	–	5119	–	–	–	1013	0
Case 1.2 – Reserve Margin	–	5517	2574	–	2202	–	2349	–	3270	738
Case 1.3 – Emissions & RPS Constraints	10,302	3616	13,204	14,796	–	6,16	0	12	–	–
Case 1.4 – Social Cost of CO ₂ and CH ₄ Leakage	4613	16,120	3114	–	2129	4113	2427	–	–	–
Case 1.5 – Deterministic Health Damages	9588	14,868	3001	–	848	–	2594	5279	–	–
Case 1.6 – Simulated Health Damages	28,495	2977	3016	–	3972	7518	3980	2717	–	–

further increases in combined cycle gas turbine generation, wind (on land and off-shore), biomass, and hydro generation. Coal is no longer in the dispatch plan in this scenario.

New capacity investments proposed by our GEP model scenarios are given in Table 6. The Baseline (Case 1.1) case proposes investments in natural gas, coal, combined heat and power, and petroleum capacity. Similarly, the Reserve Margin (Case 1.2) case includes investment in baseload units, with additional investments in combined cycle gas turbines. When societal damages are in the objective function, coal is no longer included in the investment portfolio (Cases 1.4 through 1.6), which is consistent with the findings, where coal generation was displaced with natural gas and combined cycle gas turbines. In the Social Cost of CO₂ and CH₄ Leakage scenario (Case 1.4), investments are dominated by natural gas units. Adding deterministic health damages (Case 1.5) yields similar results, but with increased investments in wind (on land and off-shore). The Simulated Health Damages case (Case 1.6) proposes further increases in wind capacity, which are more than double than the capacity proposed by the Deterministic Health Damages scenario. This increase in wind capacity proposed by the Simulated Health Damages case brings the total wind capacity of

the network to over 33 GW by 2040, which is a significant increase over the initial network capacity of approximately 2 GW, translating to over 10% of the total electricity generation in this region. Since this is a significant capacity increase, further research is required to fully assess the challenges and feasibility of the wind capacity expansions proposed in the Simulated Health Damages case.

Table 7 displays the CO₂, NO_x, and SO₂ emissions for each of the GEP scenarios. Generally speaking, the emissions output follows the corresponding dispatch plans, since displacing coal with other technologies leads to proportionate emissions reductions relative to the Baseline scenario. The inclusion of emissions limits and renewable portfolio standards as GEP constraints (Case 1.3) gives the largest emissions reductions in market cost models; however, when health and other damages are included in the objective function, we observe significant reductions in CO₂, NO_x, and SO₂ emissions relative to the market cost models (Cases 1.4 through 1.6).

Fig. 4 displays the total societal costs for each of the six cases presented. The Baseline, Reserve Margin, and Emissions Limits and RPS cases are market cost models (Cases 1.1 through 1.3

Table 7
Emissions summary.

	CO ₂ Emissions (Short Tons, Millions)	NO _x Emissions (Short Tons, Thousands)	SO ₂ Emissions (Short Tons, Thousands)
Case 1.1 – Baseline	8466	20,733	39,919
Case 1.2 – Reserve Margin	8125	19,443	36,101
Case 1.3 – Emissions Limits & RPS Constraints	6010	13,398	21,242
Case 1.4 – Social Cost of CO ₂ and CH ₄ Leakage	4473	8415	4870
Case 1.5 – Deterministic Health Damages	4114	8390	5972
Case 1.6 – Simulated Health Damages	2812	7136	7611

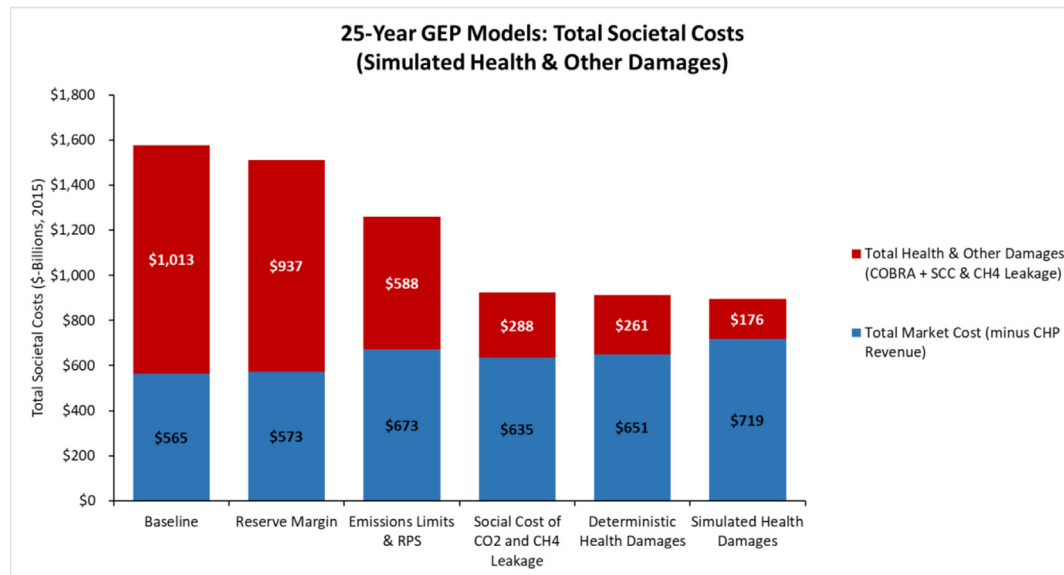


Fig. 4. Total societal cost summary (including simulated health damages).

respectively); whereas, the remainder of the cases are societal cost models (Cases 1.4 through 1.6). In comparing the market cost models (Cases 1.1 through 1.3) to the societal cost models (Cases 1.4 through 1.6), the health damages are significantly greater than in cases where total societal costs are optimized. Furthermore, total market costs for Case 1.6 are more expensive than the market cost models. Since market cost models only consider variable, fixed, and investment costs in their objective functions, this yields more capacity investments in fossil fuels and corresponding increases generation from these sources. However, in societal cost models, in particular Case 1.6, generation from fossil fuels sources are displaced in favor of low emitting energy sources. This is due to the inclusion of include health and other damages in the objective function of the societal cost models.

Based on the market cost summary in Fig. 5, as additional constraints or objective function costs are added to the GEP model,

it drives increased investment and variable O&M costs (including fuel costs) relative to the baseline. As evidenced by Case 1.3, adding emissions limits and renewable portfolio standards as model constraints drives increased investments in CHP generation to generate revenue and minimize market costs. Minimizing total societal costs ultimately leads to more expensive capacity investments, as the model selects investment options that have the least emissions, thus benefitting society from a human health perspective.

To expand on the results presented in Figs. 4 and 6 is presented to display a summary of health and other damages for each of the scenarios under investigation. As evidenced by Fig. 6, market cost minimization models yield greater health damages are greater compared to cases where total societal costs are minimized in the objective function. Since societal cost models drive significant reductions in fossil fuel generation, this reduces emissions levels, which leads to reductions in health damages.

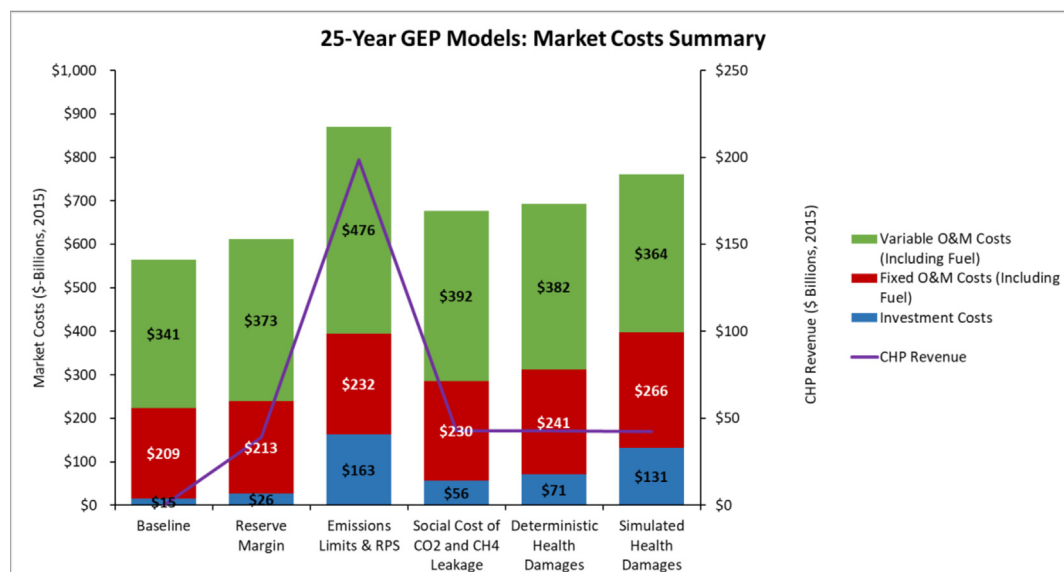


Fig. 5. Market cost summary.

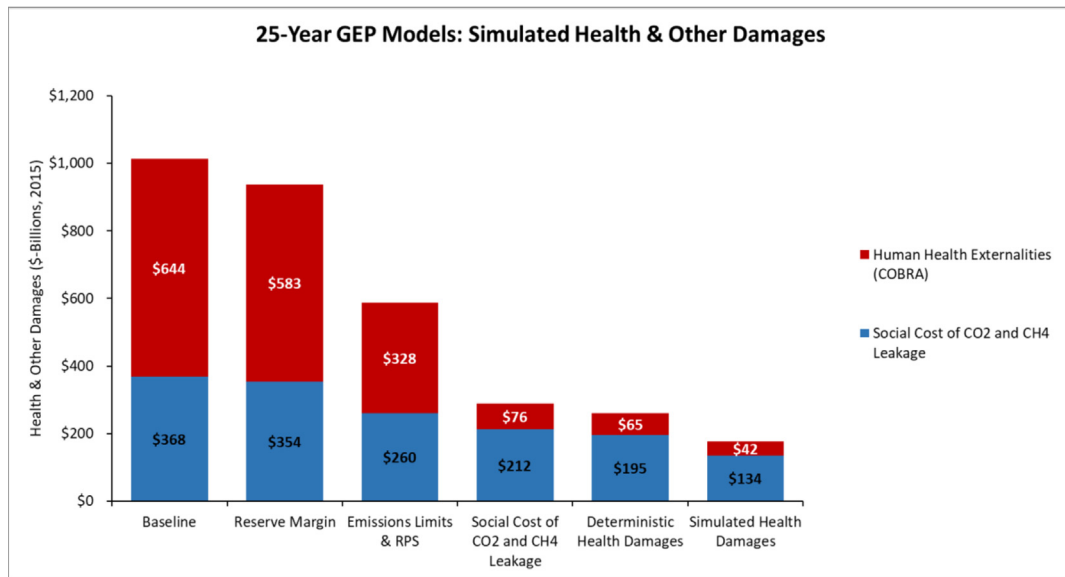


Fig. 6. Simulated health damages summary.

Case 1.6, the Simulated Health Damages case, applies the simulation-based optimization framework to solve the GEP model and obtain a capacity expansion plan that minimizes total societal costs. This procedure is summarized in Fig. 1, and described in detail in Section 2 of this paper. With each iteration, the objective function behaves erratically; however, the expected improvement value ultimately decreases to zero. As depicted in Fig. 7, the procedure is terminated after 22 full iterations, as the expected improvement value decreases and eventually reaches zero.

3.1. Discussion

Using the methodology presented in Section 2 of this paper, a multi-period, total societal cost minimization GEP problem is formulated and solved using a simulation-based optimization approach. In comparing the results of this approach to various cases, the proposed method is more effective at mitigating human

health damages associated with power grid capacity expansion decisions, than leveraging other alternatives, such as using deterministic penalties in the objective function or implementing emissions limits and renewable portfolio standards as model constraints. However, in order to realize the minimized health damages, increased investments in renewable energy sources must be made, thus driving increased market costs over the planning horizon.

While the results presented in this section are aligned with related research on this topic, as presented in Section 1, this presents a potential feasibility challenge due to large investments in wind technology as an energy source. The proposed wind capacity expansions, as outlined in the Simulated Health Damages, case would increase the total wind capacity of the network to over 33 GW by 2040. This capacity expansion, from an initial network capacity of approximately 2 GW, translates to over 10% of the total electricity generation in region under investigation. This is in line

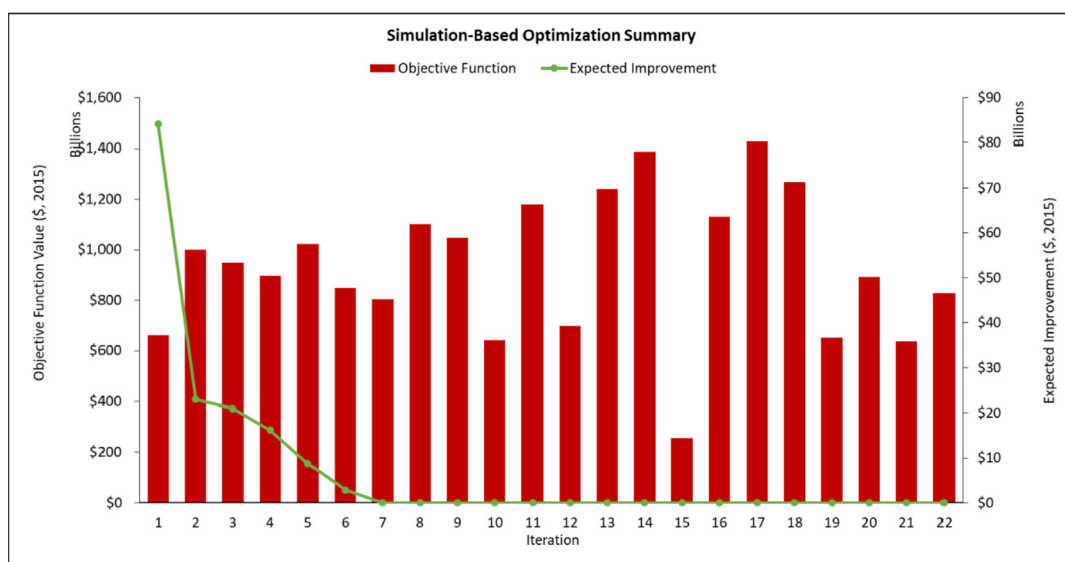


Fig. 7. Simulation-based optimization iteration summary.

with a 2008 United States Department of Energy study that proposes a pathway to satisfy 20% of the U.S. demand for electricity using wind energy by 2030. To achieve this goal, this requires approximately 300 GW of additional wind capacity, 100 GW of which are made in the Northeastern U.S. [41]. While the Department of Energy's research suggests that the proposed wind capacity expansions, while much more aggressive than the expansion plan proposed in this paper, are feasible, further research is required to fully assess the challenges of this investment strategy.

This research enables policy makers to make more informed decisions on expanding power grid capacity. Since typical GEP models do not include health damages in the objective function, the resulting expansion plans include fossil fuel investments. While these energy sources can economically (in terms of market costs) and reliably satisfy the load on the system, they also produce emissions that are potential health hazards. This research presents a framework for decision makers to mitigate these concerns, while understanding the resulting implications of a given capacity investment strategy.

4. Conclusions

This paper presents an optimization framework to solve the GEP problem that minimizes total societal damages, inclusive of market costs (investment costs, fixed and variable operating & maintenance costs, and fuel costs) and health and other damages associated with pollutant emissions from electricity generation via simulation-based optimization methods. Traditional approaches to mitigating health and other damages in the context of GEP, such as including emissions limits and renewable portfolio standards as model constraints or applying deterministic multipliers as penalties in the objective function, are not as effective at reducing the health effects from electricity generation in comparison to the analytical framework presented in this research. This is the first comprehensive attempt at using advanced statistical and simulation methods to quantify the health damages in the context of GEP. This allows researchers and policy makers to make more informed decisions on expanding power grid capacity.

Since the models propose significant investments in renewable energy sources, this work may be extended to include elements of intermittency and disruption in their supply. To assess this issue, we may consider the following optimization approaches (amongst many other potential variations): (i) assuming the capacity factors of these units to be stochastic parameters in our model, (ii) leveraging simulation to estimate the capacity factors of renewable units, or (iii) applying seasonality factors to the capacity factors. Since our model proposes significant wind capacity expansions, these research extensions are of major relevance, as further research is required to fully address feasibility concerns along other challenges associated with wind capacity expansions.

In conclusion, this research makes the following research contributions: (i) employing a kriging metamodel to express the relationship between GEP decisions and health damages, (ii) solving a large-scale, multi-period optimization model over an extended time horizon with this metamodel as a surrogate objective function, and (iii) developing an iterative procedure to resample and refit the metamodel to yield robust candidate solutions.

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